Yijie Ma

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Skills

C/C++, Go, Python, Networking, Distributed Systems, Kubernetes, OpenTelemetry, Storage Systems, Cross-platform (Linux/macOS/Windows) software development

Work Experience

Google — April 2017 - Present

Software Engineer

Google Cloud API Gateway & Service Infrastructure

- Built C++ OpenTelemetry distributed tracing across HTTP and gRPC for Cloud, GenAI, and MCP APIs.
- Built W3C Trace Context propagation, caller identity attribution, and Agent-to-Agent (A2A) telemetry.
- Led the architecture and launch of a decoupled, fast-track API configuration rollout pipeline with dual-stack routing.
- Built LLM multi-agent release automation workflows to analyze code changes, assess risks, and gate rollouts.

Google Kubernetes Engine (GKE) Networking

- Built GKE control-plane automation in Go to provision multi-NIC RDMA network topologies for AI GPU clusters.
- Built concurrent provisioning pipelines and validated multi-node NVIDIA H200 RDMA (~345 GB/s NCCL).

gRPC Open Source Remote Procedure Call Framework

- Core developer and Maintainer; designed and built the cross-platform asynchronous DNS resolver.
- Designed and built the Metrics framework with C++ template technique and the OpenTelemetry plugin.
- Co-authored the A79: Non-per-call Metrics Architecture gRFC.
- Fixed numerous crashes, security issues and made performance improvements for millions of Open Source users.

Google Drive Desktop application

- Built rsync-style differential compression and Gzip compression stack for content upload.
- Built USB device backup stack which backs up millions of users' files daily.
- Built Cross-platform Preference system for users and admins to configure the application.
- Made 100+ crash fixes, customer issue fixes and performance improvements.

Stadia Cloud Gaming Service

- Built a Cgroup library for setting up Cgroup configurations during container setup on VMs.
- Increased click-to-play speed by 75% by optimizing the container setup.
- Built a custom shell to replace explorer.exe for gamer surface lockdown on Windows.

NetApp — August 2015 - March 2017

Member of Technical Staff

Worked on the AltaVault Cloud-Integrated Storage device. Built a filesystem software and used RocksDB for cataloging users' snapshot data. Built full or incremental backup and restore functionalities.

Education

Carnegie Mellon University — Sep 2013 - May 2015

Master of Science in Electrical and Computer Engineering

Coursework: Introduction to Computer System, Embedded Realtime System, Advanced Storage System, Internet Service, Cloud Computing, Distributed System

Donghua University — Sep 2009 - May 2013

Bachelor of Engineering in Electrical Engineering and Automation

Awards: 2012 IEEE-standard Micromouse competition national award issued by China Computer Federation (CCF)